

Self-Adaptive Q-Learning for Multipath Routing in Dynamic UAV Ad Hoc Networks

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Abstract. Routing in UAV networks like FANETs is a major challenge due to high mobility, rapidly changing topology, and unstable wireless links. Classical protocols like AODV and DSR use single-path routing, making them vulnerable to link disruption. Multipath methods like AOMDV or PMAODV improve reliability by maintaining multiple alternative routes, but still rely on static criteria. Recently, reinforcement learning based approaches, from Q-routing to Q-learning, DRL, or multi-agent methods, allow network nodes to learn forwarding policies from local observation. However, in most studies, the learning parameters remain fixed, limiting adaptability. This paper proposes SA-QMAODV, a Self-Adaptive Q-learning-based Multipath routing protocol. Specifically, the method incorporates an event-driven discovery mechanism for link failures, adaptive learning speed based on topological changes, and a context-aware reward function that combines reliability, latency, and energy. This allows nodes to flexibly balance discovery and exploitation while adapting well to network changes. Simulation results on NS-3 with cloud infrastructure show that SA-QMAODV significantly improves packet delivery rate and latency compared to representative protocols, while the additional cost is negligible. This demonstrates the potential of adaptive learning mechanisms in routing for dynamic, infrastructure-less IoT networks.

Keywords: UAV-enabled ad hoc networks; Multipath routing; Reinforcement learning; Q-learning; Adaptive routing; Edge intelligence; Network performance optimization

1 Introduction

The Internet of Things (IoT) development has spurred many distributed applications such as environmental monitoring, disaster response, and smart agriculture. In infrastructure-less environments, maintaining connectivity to a centralized network is not feasible. In this context, Unmanned Aerial Vehicles (UAVs) provide a flexible solution by forming Flying Ad-hoc Networks (FANETs), enabling multi-hop communication without the need for fixed infrastructure [10]. However, FANETs are highly dynamic due to the movement of nodes, constant topological changes, and unstable links, making routing difficult [9].

Traditional routing protocols such as AODV [1] use single-path model, resulting in frequent path breaks and high latency in dynamic environments [10]. Multipath extensions such as AOMDV [4] and PMAODV [5] improve reliability by maintaining multiple routes, with PMAODV using a probabilistic forwarding mechanism. However, these methods still rely on static decision mechanisms and lack adaptability to real-time network conditions.

To increase adaptability, reinforcement learning (RL) has been introduced into the routing problem. Early studies such as Q-routing [6] and later methods such as QMR [7], QLAODV [8], RL-AODV [11] demonstrate the potential of Q-learning in dynamic environments. In FANETs, RL-based methods further improve performance by allowing nodes to learn from local observations [12], [13]. In particular, QMAODV [17] combines Q-learning with multipath routing, allowing next-node selection based on latency and ACK responses.

However, most RL-based routing protocols rely on fixed learning parameters, such as exploration rate, learning rate, and reward weights, limiting adaptability under varying network conditions [18], [19]. This results in suboptimal performance in highly dynamic environments.

To address this limitation, this paper proposes SA-QMAODV, a Self-Adaptive Q-learning-based Multipath routing protocol for UAV-enabled networks. The proposed approach extends QMAODV by introducing adaptive mechanisms for exploration, learning rate, and reward weighting, enabling nodes to dynamically adjust their learning behavior based on real-time conditions. In particular, the design incorporates an event-driven exploration strategy, a topology-aware learning rate, and a context-aware reward function [21].

The protocol is implemented in NS-3 and evaluated using a cloud-based simulation framework with parallel execution. Experimental results show that SA-QMAODV significantly improves packet delivery ratio, end-to-end delay, and maintains stable throughput compared to AODV, PMAODV, and QMAODV, with minimal overhead.

2 Related Work

Routing in Mobile Ad-hoc Networks (MANETs) and UAV networks has been extensively studied, with numerous protocols proposed to address challenges such as constantly changing topology, limited bandwidth, and the lack of a fixed infrastructure. Initial solutions are generally divided into two main groups: proactive and reactive. Proactive protocols such as DSDV [3] maintain routing tables through periodic updates, enabling low-latency forwarding but at the cost of high control overhead. Conversely, reactive protocols such as AODV [1] and DSR [2] only establish routes when needed, reducing overhead but increasing latency during path discovery. While performing well in low to medium mobility environments, these protocols often struggle in highly dynamic scenarios such as UAV networks, where routes are frequently disrupted [10].

To improve reliability and load balancing, multipath routing protocols have been developed. A typical example is AOMDV [4], which extends from AODV by maintaining multiple non-loop routes and separating links to the same destination. As

a result, AOMDV provides alternative routes, reducing the need to rediscover the route when errors occur. Based on this idea, PMAODV [5] further proposes a probabilistic forwarding mechanism, distributing traffic across multiple paths based on probabilities derived from the hop count. This approach improves load balancing and fault tolerance compared to single-path routing. However, in both AOMDV and PMAODV, the decision-making process remains static, relying on predefined metrics instead of reflecting real-time network conditions.

To overcome the limitations of static routing strategies, many recent studies have applied RL to the routing problem. One of the early works was Q-routing [6], which uses Q-learning to estimate the expected delay to the destination through neighboring nodes. This approach has opened up many learning-based routing protocols, such as QLAODV [8] and RL-AODV [11], in which nodes learn to choose the optimal hop based on local observations such as delay, packet transfer success rate or link quality. In the context of UAV and FANET networks, studies have also shown that RL can improve routing performance in dynamic environments by allowing nodes to adapt to network changes without needing global information [7], [12], [13].

Despite significant progress, enhancement-based routing protocols still have some notable limitations. First, many methods still rely on single-path routing, limiting the ability to exploit path diversity and reducing resilience to link failures. Second, most protocols use fixed learning parameters such as discovery rate, learning rate, and reward weights, which are often selected empirically and do not change throughout operation. This limitation has been noted in many recent studies [18], [19], where static configuration reduces adaptability in highly variable network conditions. Third, the number of studies focusing on distributed, lightweight adaptation mechanisms suitable for resource-constrained UAV nodes is still quite small [20].

Recently, some studies have begun to combine enhancement learning with multipath routing to improve adaptability in dynamic networks. For example, QL-AODV [22] is proposed for UAV networks geared towards 6G, where Q-learning is integrated to optimize route selection in highly mobile environments. Similarly, QLA-MAODV [23] extends this approach to the multicast routing problem, allowing for adjustment of routing behavior based on feedback from the network. These works show the clear potential of combining RL with multipath routing. However, most methods still use fixed learning parameters or only adapt to a limited extent, thus not fully exploiting the self-adjustment capabilities of the learning process in rapidly changing network conditions.

In summary, although previous studies have considered multipath routing and RL separately, there remains a clear gap in combining these two approaches with adaptive learning mechanisms. To our knowledge, no existing work simultaneously integrates multipath routing, distributed Q-learning, and real-time parameter tuning within a unified framework. This gap is the impetus for proposing the SA-QMAODV protocol, which aims to provide a highly reliable, adaptive, learning-based routing mechanism for infrastructureless UAV networks.

3 Background: QMAODV Overview

The QMAODV protocol is an extension of AODV, in which RL is integrated into the routing decision-making process, while retaining the advantages of multipath routing. Inheriting ideas from protocols such as AOMDV and PMAODV, QMAODV allows each node to maintain multiple candidate paths to the same destination. Instead of using only a single optimal route, each node stores a set of next-hops obtained from the route discovery and feedback process. This multipath structure increases fault tolerance and reduces the frequency of route rediscovery when link loss or congestion occurs.

In principle, QMAODV models the routing problem as a Q-learning-based RL process [6]. Each node acts as an agent, learning to select the optimal next hop through interaction with the network environment. The state is defined as the destination node, while the action is the selection of the next hop from available candidates. After each packet forwarding, the node receives feedback from the network in the form of an ACK signal and transmission delay, from which it calculates the reward. The value of $Q(s, a)$ is updated to estimate the long-term benefit of selecting the next hop a when sending to destination s . This process is repeated continuously, allowing nodes to gradually learn the optimal routing policy based on accumulated experience.

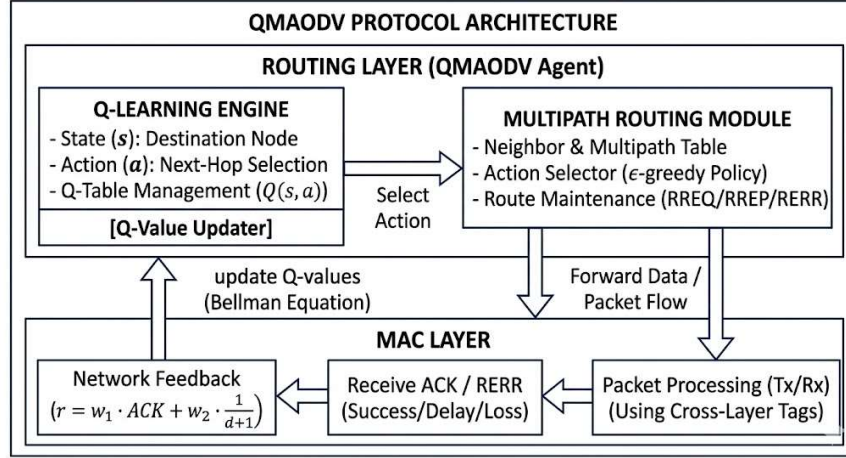


Fig. 1. QMAODV Architecture

The QMAODV architecture is illustrated in Fig. 1, where each node maintains a Q-table mapping between the pair (destination, next-hop) and the corresponding Q value. Routing decisions are made according to a balanced strategy between exploration and exploitation, often using the ϵ -greedy policy [11]. This allows the node to both experiment with new routes (exploration) and utilize well-evaluated routes (exploitation). This approach allows QMAODV to adapt better than traditional static routing protocols.

However, despite incorporating learning into the routing process, QMAODV still has some important limitations. Specifically, learning parameters such as the learning rate α , the discount factor γ , and the discovery rate ε are often fixed. Similarly, the reward function uses static weights for components such as ACK and latency. This configuration is common in many RL-based routing protocols [11], [18], where parameters are manually fine-tuned and do not change throughout operation. Therefore, the system struggles to adapt to constantly changing network conditions, such as fluctuations in travel speed, node density, or link quality. This can lead to suboptimal routing decisions in highly dynamic environments such as FANETs [9].

These limitations highlight the need for a more flexible and adaptive learning mechanism. Recent studies have highlighted the importance of dynamically tuning learning parameters and reward functions contextually [19], [21]. This is the impetus for proposing SA-QMAODV, a protocol that extends QMAODV with self-adaptive mechanisms to enhance learning efficiency and improve routing performance in dynamic UAV networks.

4 Proposed Method: SA-QMAODV

4.1 Design Overview

The proposed Self-Adaptive Q-Learning Multipath AODV (SA-QMAODV) protocol extends the QMAODV framework by incorporating a dynamic learning mechanism that enables nodes to autonomously adjust their routing behavior based on real-time network conditions. The design of SA-QMAODV follows an incremental approach, preserving the strengths of existing multipath and learning-based routing while addressing their key limitations through adaptive enhancements.

First, SA-QMAODV retains the multipath routing structure inherited from PMAODV [5]. During the route discovery phase, multiple candidate paths to a destination are established and maintained in the routing table, similar to AOMDV [4]. Each path corresponds to a potential next-hop neighbor, providing redundancy and enabling load balancing. This multipath capability improves fault tolerance and reduces the frequency of route rediscovery in highly dynamic environments such as UAV-enabled networks [10].

Second, SA-QMAODV builds upon the Q-learning model introduced in QMAODV [17] and inspired by classical Q-routing approaches [6]. Each node maintains a local Q-table that maps destination–next-hop pairs to Q-values, representing the expected long-term utility of forwarding packets through a given neighbor. Routing decisions are made using an exploration–exploitation strategy, such as ε -greedy, and Q-values are updated continuously based on feedback such as acknowledgment success and transmission delay [8], [11]. This decentralized learning approach allows nodes to adapt their forwarding strategies without requiring global network knowledge or centralized coordination, which is particularly suitable for highly dynamic FANET environments [9].

The core innovation of SA-QMAODV lies in the addition of an adaptive learning layer on top of the traditional Q-learning framework. Unlike QMAODV, where

learning parameters are kept fixed [17], SA-QMAODV allows dynamic adjustment of important parameters such as discovery rate ϵ , learning rate α , and reward function weights during operation. This approach is driven by recent studies on adaptive RL in wireless networks [19], [21], which show that static parameters often degrade performance in highly non-stop network environments. Thanks to the self-adapting mechanism, each node can adjust its learning behavior in real time, responding more effectively to fluctuations in mobility, congestion, and link stability.

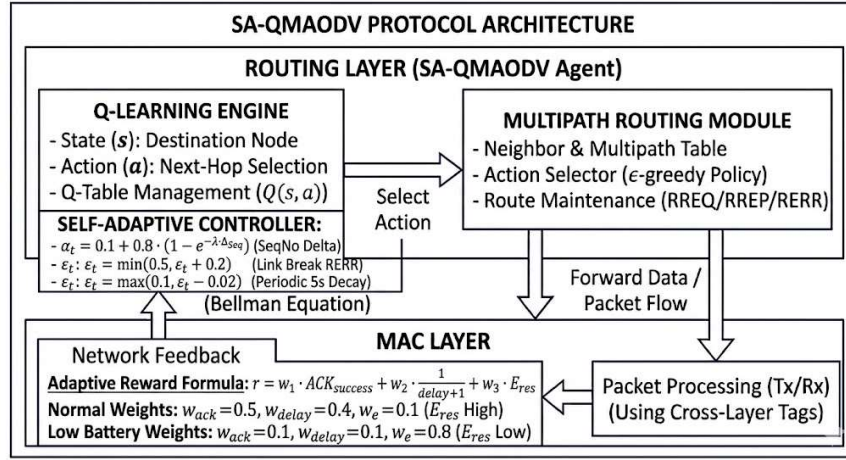


Fig. 2. Architecture of SA-QMAODV with adaptive learning layer integrated into multipath Q-learning routing.

As illustrated in Fig. 2, the architecture of SA-QMAODV is organized into three main components that interact closely. First is the multipath routing layer, responsible for discovering and maintaining a set of candidate routes to the destination, similar to the AOMDV/PMAODV protocols. Second is the Q-learning engine, which performs next-hop selection based on the ϵ -greedy policy and updates the Q value based on network feedback. Third, and a crucial difference, is the adaptive control module. This module continuously monitors local network signals such as link loss events (RERR), latency fluctuations, or sequence number changes (SeqNo), thereby dynamically adjusting the learning parameters. Specifically, when an unstable network is detected, the module increases ϵ to promote discovery and increases α to speed up adaptation; conversely, under stable conditions, these parameters are gradually reduced to ensure convergence and avoid oscillations. Simultaneously, the weights in the reward function are adjusted to balance reliability, latency, and power consumption.

This layered design allows SA-QMAODV to easily integrate with existing AODV frameworks, while adding only minimal computational and control signal overhead. As a result, the protocol not only retains the advantages of multipath routing and enhancement learning, but also significantly improves adaptability.

The integration of a self-adapting mechanism helps SA-QMAODV improve both the responsiveness and stability of the learning process. This leads to more flexible and efficient routing decisions in highly dynamic FANET environments where network conditions change constantly over time.

4.2 Adaptive Exploration Strategy

In QMAODV, the exploration rate ε is typically fixed or follows a predefined decay schedule [17], which does not adequately capture the highly dynamic nature of UAV-enabled ad hoc networks. In such environments, topology changes and frequent link failures can render static exploration strategies ineffective. A low ε may hinder route discovery after topology changes, while a high ε may reduce routing stability.

To address this limitation, SA-QMAODV adopts an event-driven adaptive exploration mechanism. The exploration rate at time t , denoted as ε_t , is dynamically adjusted using both reactive increase and periodic decay:

- Link failure-triggered increase (re-exploration): $\varepsilon_t = \min(0.5, \varepsilon_t + 0.2)$
- Periodic decay (stabilization): $\varepsilon_t = \max(0.1, \varepsilon_t - 0.02)$

When a route error (RERR) occurs, ε_t is increased to encourage exploration of alternative paths. Conversely, ε_t is gradually reduced over time to stabilize routing decisions under steady conditions. This mechanism can be viewed as a lightweight realization of $\varepsilon_t = f(\text{network stability})$, where instability is implicitly captured through routing events rather than explicit metrics such as delay variance or PDR [21].

Compared to metric-based approaches, the proposed strategy leverages existing control signals, making it efficient and suitable for resource-constrained UAV nodes. As a result, SA-QMAODV dynamically balances exploration and exploitation: increasing exploration under unstable conditions and favoring exploitation when the network stabilizes. This significantly improves routing adaptability and robustness compared to fixed-parameter RL-based protocols [9], [19].

4.3 Adaptive Learning Rate

In conventional Q-learning-based routing protocols such as QMAODV, the learning rate α is typically configured as a fixed parameter [17]. This parameter determines the extent to which newly observed information influences the existing Q-values. While simple to implement, a fixed learning rate is not suitable for highly dynamic environments such as UAV-enabled ad hoc networks. A large α may lead to unstable updates and oscillations under noisy conditions, whereas a small α may slow down the learning process, preventing timely adaptation to topology changes and link variations.

To resolve this issue, SA-QMAODV introduces an adaptive learning rate mechanism, where the learning rate at time t , denoted as α_t , is dynamically adjusted based on real-time network dynamics. Instead of relying on static configuration, α_t is defined as a function of topology variation, which is estimated using routing-level indicators.

In this study, the dynamic level of the network is quantified through the variation of the destination sequence number (SeqNo), an important parameter in AODV-based protocols reflecting network topology changes [1]. The frequent updating of SeqNo indicates that routing information is changing rapidly, corresponding to a highly dynamic network environment.

To adapt to the non-stationary nature of the environment, the learning rate (1) is designed as a monotonic function of the variation of SeqNo:

$$\alpha_t = 0.1 + 0.8 \cdot (1 - e^{-\lambda \cdot \Delta_{Seq}}) \quad (1)$$

Where Δ_{Seq} is the number of SeqNo updates observed in a short time window, and λ is the sensitivity adjustment coefficient of the adaptation mechanism.

This function is built on two main bases. Firstly, in RL, when the environment changes rapidly, a high learning rate is needed to enable the agent to adapt quickly to new conditions; conversely, when the environment is stable, a low learning rate helps ensure stable convergence. Secondly, the exponential form is chosen because it provides smooth and bounded transitions, avoiding sudden changes in parameters that can destabilize the learning process.

Specifically, formula (1) satisfies the following properties:

- Monotonously increasing with Δ_{Seq} , ensuring that the more dynamic the network, the greater the learning rate.
- Bounded in the range $[0.1, 0.9]$, consistent with the stable values commonly used in Q-learning.
- Rapidly increasing in the initial stage and gradually saturating, allowing for a quick response to initial fluctuations but limiting oscillations when the level of fluctuation is too large.

Therefore, when the network is highly dynamic (large Δ_{Seq} value), α_t approaches 0.9, helping the learning process to adapt quickly. Conversely, when the network is stable (small Δ_{Seq} value), α_t approaches 0.1, ensuring stability and convergence.

Although the proposed formula is heuristic, it is built on the fundamental principles of learning in a non-stationary environment and is validated through simulation, showing a clear effectiveness in improving routing performance in highly dynamic UAV networks.

In addition to topology-based adaptation, the learning rate can also be interpreted through the lens of the temporal-difference (TD) error, a common measure in RL [6]. A large TD-error indicates that the current Q-values are inaccurate, suggesting the need for faster updates (higher α_t), whereas a small TD-error implies convergence, favoring a smaller learning rate. Although TD-error is not explicitly used in the proposed implementation to minimize computational overhead, the SeqNo-based formulation implicitly captures similar dynamics in a lightweight and distributed manner.

By dynamically adjusting the learning rate, SA-QMAODV achieves an effective balance between responsiveness and stability. Nodes can rapidly incorporate new information when the network changes while maintaining stable routing behavior under steady conditions. This adaptive mechanism significantly improves the efficiency and

robustness of the learning process compared to fixed-parameter approaches in existing RL-based routing protocols [9], [19].

4.4 Adaptive Reward Function

In QMAODV, the reward function (2) plays a important role in guiding the learning process by evaluating the quality of each forwarding decision. The original design employs a fixed weighted combination of acknowledgment success and transmission delay, where the contribution of each metric is determined by predefined constant coefficients. While this formulation is effective under relatively stable conditions, it lacks the flexibility required for highly dynamic environments such as UAV-enabled FANETs. In practice, network conditions may change rapidly due to node mobility, congestion, and varying link quality, making static reward configurations insufficient for capturing real-time routing priorities.

To address this limitation, SA-QMAODV introduces an adaptive reward function in which the contribution of each performance metric is dynamically adjusted based on current network conditions and node states. The reward at time step t is defined as:

$$r_t = w_1(t) \cdot ACK_{success} + w_2(t) \cdot \frac{1}{delay + 1} + w_3(t) \cdot Energy \quad (2)$$

where $w_1(t)$, $w_2(t)$, and $w_3(t)$ represent time-varying weights associated with reliability, latency, and energy-awareness, respectively, subject to the normalization constraint:

$$w_1(t) + w_2(t) + w_3(t) = 1 \quad (3)$$

The adaptive mechanism adjusts reward weights based on local observations of each node. When network congestion is detected, evidenced by high average latency or latency variance, the weight of the latency component $w_2(t)$ will be increased to prioritize routes with lower latency. Conversely, when a poor link is detected through high packet loss rate or frequent ACK failures, the reliability weight $w_1(t)$ will be increased to favor more stable paths.

In addition to network-level factors, SA-QMAODV also integrates node-level awareness through an adaptive energy mechanism. Specifically, when a UAV has low remaining energy (e.g., below 20%), the energy weight $w_3(t)$ will be significantly increased. This reduces the probability of selecting low-energy nodes as relays, thereby helping to distribute the load more efficiently and extend the overall network lifespan.

Thanks to this design, the reward function can flexibly shift the optimal goal between reliability, latency, and energy efficiency depending on the actual state of the network. Compared to static reward functions, the proposed mechanism significantly improves the adaptability and robustness of the learning process, making routing decisions more context-appropriate in a highly dynamic FANET environment.

To summarize the enhancement learning configuration in SA-QMAODV, Table 1 presents the main parameters and their corresponding adaptive characteristics.

Table 1. Reinforcement Learning Parameters for SA-QMAODV

Parameter	Value
Learning rate α_t	$\alpha_t = 0.1 + 0.8 \cdot (1 - e^{-\lambda \cdot \Delta_{Seq}}), \alpha_t \in [0.1, 0.9]$
Discount factor γ	$\gamma = 0.9$
Exploration rate ε_t	$\varepsilon_t = \begin{cases} \min(0.5, \varepsilon_t + 0.2), & \text{upon RERR} \\ \max(0.1, \varepsilon_t - 0.02), & \text{periodically} \end{cases}$
Initial Q-values	$Q(s, a_i) = \frac{1/HC_i}{\sum_{k \in A(s)} (1/HC_k)}$
Reward function	$r_t = w_1(t) \cdot ACK_{success} + w_2(t) \cdot \frac{1}{delay + 1} + w_3(t) \cdot Energy$
Reward weights	$\begin{cases} \text{Normal: } (w_1 = 0.5, w_2 = 0.4, w_3 = 0.1) \\ \text{Low - energy: } (w_1 = 0.1, w_2 = 0.1, w_3 = 0.8) \end{cases}$

This multi-dimensional adaptive reward design allows SA-QMAODV to jointly optimize reliability, latency, and energy efficiency, which is particularly critical for resource-constrained and highly dynamic UAV-enabled networks.

4.5 Updated Q-learning Rule

To incorporate the proposed adaptive mechanisms into the learning process, the standard Q-learning update rule is extended by introducing time-varying parameters. In particular, both the learning rate and the reward are dynamically adjusted at each time step, resulting in the following update formulation:

$$Q(s, a) \leftarrow (1 - \alpha_t) Q(s, a) + \alpha_t \left[r_t + \gamma \max_{a'} Q(s, a') \right] \quad (4)$$

where α_t denotes the adaptive learning rate defined in Section 4.3, and r_t represents the adaptive reward described in Section 4.4. The discount factor γ remains constant to ensure stability in long-term reward estimation.

By incorporating α_t and r_t , the learning process becomes more responsive to dynamic network conditions. A larger value of α_t enables faster adaptation when significant topology changes occur, whereas a smaller value helps stabilize the learning process under relatively steady conditions. Meanwhile, the adaptive reward function allows the routing agent to dynamically prioritize different performance metrics, such as reliability, delay, and energy efficiency, depending on the current state of the network. This extended formulation enables the Q-learning agent to continuously refine its routing policy in a context-aware manner, thereby improving convergence speed and decision quality in highly dynamic UAV-enabled ad hoc networks.

4.6 Algorithm

The overall operation of the proposed SA-QMAODV protocol extends the QMAODV workflow by incorporating adaptive learning components during packet forwarding.

The route discovery and maintenance procedures remain consistent with AODV, PMAODV, while the learning-enhanced forwarding process is modified as follows.

Algorithm 1: SA-QMAODV Operation at Each Node

Input: Destination s , available next-hop set $A(s) = \{a_1, a_2, \dots, a_n\}$
Output: Selected next-hop and updated Q-values

- 1. Initialization:**
 - Initialize Q-table using normalized inverse hop count
 - Set initial parameters: $\alpha_0, \gamma, \epsilon_0$
 - Initialize reward weights $w_1(0), w_2(0), w_3(0)$
- 2. Route Discovery (unchanged):**
 - If no route to destination S :
 - Broadcast RREQ
 - Receive RREP(S), establish multipath routes
 - Initialize Q-values for each next-hop
- 3. Packet Forwarding Loop:**
 - For each packet destined to s :
 - Step 1: Compute adaptive parameters**
 - Estimate network conditions (e.g., delay variance, ACK success rate)
 - Compute exploration rate ϵ_t (Section 4.2)
 - Compute learning rate α_t (Section 4.3)
 - Update reward weights $w_1(t), w_2(t), w_3(t)$ (Section 4.4)
 - Step 2: Action selection (ϵ -greedy)**
 - With probability ϵ_t : select random next-hop $a \in A(s)$
 - Otherwise: select $a = \arg \max Q(s, a)$
 - Step 3: Packet transmission**
 - Forward packet to selected next-hop a
 - Step 4: Feedback observation**
 - Measure one-hop delay
 - Detect ACK success/failure
 - Step 5: Reward computation**
 - Compute adaptive reward:
$$r_t = w_1(t) \cdot ACK_{success} + w_2(t) \cdot \frac{1}{delay + 1} + w_3(t) \cdot Energy$$
 - Step 6: Q-value update**
 - Update:
$$Q(s, a) \leftarrow (1 - \alpha_t)Q(s, a) + \alpha_t \left[r_t + \gamma \max_{a'} Q(s, a') \right]$$
- 4. Loop continues** for all packets and time steps

This algorithm enables each node to adapt its routing behavior dynamically based on real-time network feedback. By combining multipath routing with self-adaptive RL, SA-QMAODV achieves improved robustness, faster convergence, and better performance in highly dynamic UAV networks.

5 Simulation Setup

To evaluate the performance of the proposed SA-QMAODV protocol, simulations were conducted using the NS-3 network simulator under a UAV-enabled FANET scenario.

The simulation framework integrates custom implementations of AOMDV, PMAODV, QMAODV, and SA-QMAODV to ensure a consistent and fair comparison. A cloud-based execution platform using Google Cloud Run Jobs is employed to enable scalable and parallel simulations. This setup supports efficient Monte Carlo experimentation by distributing simulation tasks across multiple instances and aggregating results automatically.

The network configuration is designed to reflect a dynamic UAV environment. Simulations are executed for 200 seconds in a 500×500 m² area with 10 to 30 mobile UAV nodes. Nodes follow a random mobility model, communicating over IEEE 802.11 using Constant Bit Rate (CBR) traffic with 512-byte packets at 150 kbps. A DropTail queue model with limited buffer size is used.

To ensure statistical reliability, each network configuration (node size from 10 to 30) is repeated 30 times with different random seeds, resulting in 630 runs per protocol. Overall, more than 3000 simulation runs are executed across all protocols using parallel cloud-based infrastructure. The reported results are averaged over all runs to reduce randomness and ensure statistical stability.

The proposed SA-QMAODV protocol is compared against representative routing schemes, including AODV (single-path), AOMDV (multipath), PMAODV (probabilistic multipath), and QMAODV (learning-based with fixed parameters). This comparison enables a comprehensive evaluation across traditional, multipath, and RL-based routing approaches.

6 Results and Discussion

This section evaluates the performance of SA-QMAODV against AODV, PMAODV, and QMAODV in scenarios with different network scales. The two main metrics used are packet delivery rate and end-to-end latency, reflecting routing reliability and efficiency in a highly dynamic FANET environment.

As shown in Fig. 3, SA-QMAODV achieved the highest PDR in most cases. In low-density networks, 10 to 15 nodes, the improvement wasn't significant due to the relatively stable network structure. However, as the number of nodes increased, the network became more volatile, and performance differences became increasingly apparent. AODV had the lowest PDR due to its reliance on a single path, making it vulnerable to link failures. PMAODV improved durability through multipath but was still limited by its static probabilistic forwarding mechanism. QMAODV leverages machine learning to improve performance, but its fixed parameters limit its adaptability.

In contrast, SA-QMAODV achieves superior PDR, especially in high-density scenarios, 20 to 30 nodes, thanks to its dynamic adjustment of discovery and learning behavior. The event-driven discovery mechanism quickly finds alternative paths when

link failures occur, while the adaptive reward prioritizes stable routes, thereby improving packet delivery under unstable network conditions.

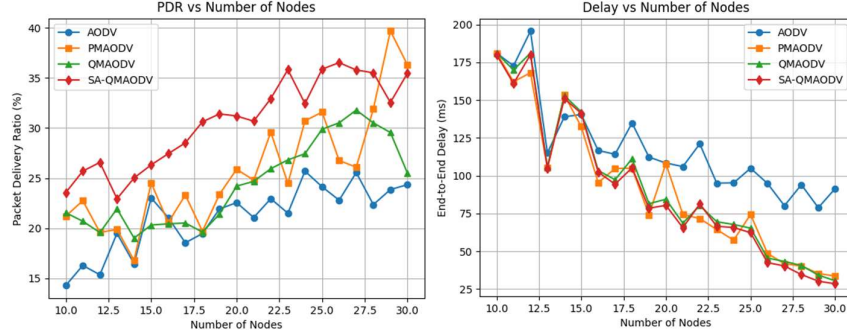


Fig. 3. Packet Delivery Ratio and end-to-end delay versus number of nodes.

Fig. 3 also shows that SA-QMAODV achieves lower or comparable latency compared to other protocols, especially as network density increases. AODV has high latency due to frequent route rediscovery, while PMAODV sometimes selects suboptimal routes when congestion occurs. QMAODV improves latency through learning, but due to fixed parameters, its response to network changes is slow.

SA-QMAODV overcomes these limitations through adaptive parameter adjustment. Increasing the discovery level during link errors and adjusting the learning rate according to network state accelerates convergence and shortens recovery time. Simultaneously, the adaptive reward mechanism prioritizes low-latency routes under congestion, contributing to improved latency performance.

Overall, SA-QMAODV outperforms the control protocols in both reliability and latency. The combination of multipath routing and adaptive enhancement learning allows for efficient handling of network topology changes, with particularly pronounced benefits in high-density and highly volatile scenarios. This result demonstrates the potential of adaptive learning mechanisms in improving routing performance for infrastructureless UAV networks.

7 Conclusions

This paper presented SA-QMAODV, a self-adaptive Q-learning-based multipath routing protocol designed for UAV-enabled ad hoc networks. The proposed approach extends the QMAODV framework by introducing adaptive mechanisms that dynamically adjust key learning parameters, including the exploration rate, learning rate, and reward function, based on real-time network conditions.

By integrating multipath routing with self-adaptive RL, SA-QMAODV enables each node to autonomously balance exploration and exploitation, accelerate learning under dynamic topology changes, and make context-aware routing decisions. The adaptive reward design further allows the protocol to jointly consider reliability, delay, and

energy efficiency, which are critical factors in resource-constrained and highly dynamic FANET environments.

The protocol was implemented in the NS-3 simulator and evaluated through a cloud-based simulation platform with parallel execution capabilities. Large-scale simulation results showed that SA-QMAODV significantly improved PDR and reduced end-to-end latency compared to AODV, PMAODV, and QMAODV, especially in scenarios with medium to high node density. These results confirm the effectiveness of integrating adaptive learning mechanisms into routing protocols in highly dynamic UAV networks.

For future work, several avenues can be explored. First, functional approximation techniques (e.g., deep reinforcement learning) can be applied to address the scalability limitations of tabular Q-learning in larger networks. Second, cooperative or multi-agent learning strategies can be investigated to improve coordination among UAV nodes and reduce local decision bias. Finally, integrating prediction-based mechanisms (e.g., predicting mobility or link quality) can further enhance routing stability and proactive adaptability in highly dynamic environments.

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